

The background is a light beige color with several decorative elements. In the top right, there are three concentric circles in a light blue-grey color, with a small orange triangle and a white circle at the center. In the bottom left, there is a dark green rounded square containing a white square with a small orange dot in the center. Thin grey lines with orange dots at the corners connect these elements across the page.

PROJECT PRESENTED BY

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—Overview

From Flat Diagrams to Living Models

The core shift this project is built around: theory you can enter
and move through, not just read.

- ◎ Bridging abstract theory and **3D spatial reality**
- ✧ Moving from passive listening to **active exploration**
- ← Amplifying the faculty role — **never replacing it**
- ⇄ For students: **eliminating unnecessary cognitive load**,
not adding memorisation

—Context

Chalkboard to Spatial Cognition

Technology moved on for banking, retail, and entertainment.
Teaching didn't.



Past

Static chalkboards & imagination alone



Present

Smart boards, still passive consumption



Future

Spatial simulation, actively experienced

—Theoretical Foundation

Grounded in Research

Not opinions — established positions in education research.

"Most engineering students are visual learners, taught in a verbal, mathematical style."

Richard Felder · NC State

"Active manipulation of concepts transfers understanding faster than standard lectures."

Eric Mazur · Harvard

Reducing unnecessary cognitive load lets the brain retain complex material far better.

Cognitive Load Theory · Sweller

—Supporting Evidence

The Evidence

The numbers behind why this matters, drawn from real published research.

80%+

Graduates lacking practical readiness (Aspiring Minds)

55%

Higher STEM failure under pure lecturing (Freeman et al., PNAS 2014)

38%

Drop in dropout/backlog after interactive tools (UNESCO 2021)

16%+

Annual growth, global immersive EdTech market

—Equity and Access

Every Student, Equal Footing

A resource gap should never quietly become a capability gap.

- 🏠 Premier institutes have **labs Tier-2/3 colleges can't match**
- 🌐 A cloud platform brings the same experience to **any smartphone**
- ✨ Students can **revisit and master concepts at their own pace**, independently
- 🎯 Resource gap $\neq$ capability gap

—Institutional Alignment

An Amplifier, Not a Replacement

The constraint was
never faculty
knowledge — it was
*45 minutes and a
flat board.*

- ⬡ Faculty depth and mentorship remain the **foundation** of engineering education
- ← A teaching companion — **displays a ready 3D model at one click**, freeing time for deeper discussion

—Value Proposition

Who Benefits

A single platform, three groups who all win from it.

STUDENTS

- Genuine understanding, not rote memorising
- Lower anxiety, stronger CGPA
- Strengthens practical engineering skills

FACULTY

- A ready teaching companion
- More class time for depth
- Improves overall pass percentage & outcomes

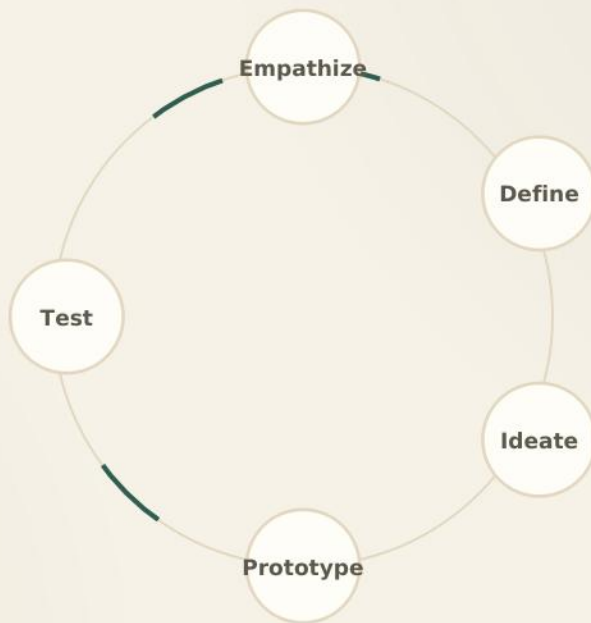
INSTITUTION

- Higher pass rates
- Strong NAAC / NBA evidence
- Reputation as a tech-forward campus

—Methodology

Our Process

Five stages, one continuous loop — empathy always comes first.



—Stage 1

Empathize

The starting point of every design-thinking process: what does the struggle actually feel like?

-  **72%** memorise formulas without understanding the mechanism
-  **85%** struggle specifically with 3D visualisation
-  **68%** cite abstract, non-visual lecturing as the main driver of backlogs
-  "I can solve it on paper — I don't know what it means in space."

—Stage 2

Define

The problem statement that everything else in this project answers to.

Students need to
interactively
manipulate
dynamic concepts
— static 2D
representation is
the barrier.

Reframed focus: not student intelligence, but delivery architecture — shifting from **memorising for exams** to **visualising to understand**.

—Stage 3

Ideate

Six features, one shared idea: let the student touch the concept, not just view it.

CORE INTERACTION

- **Multi-variable sliders** — drag mass, speed, voltage, watch the model respond
- **Layer exploder** — X-ray view into machinery, layer by layer
- **Concept-to-code** — see the geometry behind an equation

SUPPORT & SAFETY

- **"What-If" AI assistant** — instant answers, dynamically visualised
- **Faculty-guided customisation** — models mapped to the syllabus
- **Error-free experimentation** — no risk to physical lab equipment

—Competitive Positioning

Why Not Existing Tools?

Good tools already exist. Here's the gap none of them close.

PhET / GeoGebra

- Excellent, but single-subject
- Not aligned to our syllabus
- No instant doubt resolution

YouTube 3D Content

- Passive, one-way viewing
- No manipulation, no doubts answered
- Scattered across creators

EduVerse 3D

- One platform, every core subject
- Faculty-aligned, AI-assisted
- Built-in "What-If" doubt resolution

The real gap isn't better graphics — it's **one consistent interface** tied to what your own faculty is teaching that week.

—Stage 4

Prototype

System architecture and interface — every control a student needs, nothing they don't.

- 🌐 Central 3D canvas, fully rotatable, with cross-sectional view
- ☀️ Live variable controls on the side
- 📖 Explanation sidebar — theory linked live to the visual
- ✨ Dark theme, royal blue & emerald green — easy on the eyes



—Stage 4 — Applied

Magnetic Fields, Live

The same equation, taught two different ways.

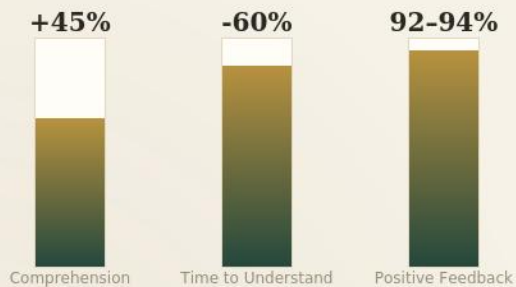
- 💡 Formula on a board → a living field in space
- 🌀 Rotate 360°, adjust time, watch it change



—Stage 5

Test

A/B tested against a control group, not just theorised.



—Evaluation

Feasible. Viable. Scalable.

Built to work on the hardware students actually own.

- ☀ Runs on **WebGL / Three.js**, cloud-rendered
- 📦 No dependence on expensive local hardware
- 📖 Same architecture spans **four+ engineering disciplines**
- 🏠 Strong, consistent demand from both **students and faculty**

—Implementation Planning

How We'd Build It

Indicative, not final — a realistic starting estimate.

STACK

- Three.js / WebGL rendering
- Progressive web app front end
- Hosted AI layer for the assistant

TIMELINE

- Core engine & one module — 6-8 weeks
- Dashboard & AI — 4-6 weeks
- Pilot with one class — 2-3 weeks

COST DRIVERS

- Cloud hosting — free tier at pilot scale
- AI API usage — scales with active students
- Biggest real cost: dev time, not infra

—Roadmap

What's Next

The core engine doesn't change — only how far its reach extends.

-  **Augmented reality** — models projected onto the desk
-  **Digital twins** of real industrial machinery
-  **AI-personalised** paths that find your weak spot
-  **Gamified peer learning** — collaborative, challenge-based modules
-  **Institutional legacy** — a home-grown solution, built by students for students

—Honest Self-Assessment

Where We're Being Honest

A proposal with no trade-offs is a proposal that hasn't been tested.

RISK

Connectivity dependence

MITIGATION

Lightweight offline mode for core models

Content-creation overhead

Launch with high-impact concepts first

Faculty adoption resistance

Start with volunteer co-designers

Device variability (low-end hardware)

Simplified rendering mode for constrained devices

—Closing

The Soul of Engineering

We are not replacing the teacher. We are strengthening the *medium* through which their knowledge reaches every mind.

True engineering is measured not by degrees or marks, but by the **perspective** with which real problems — in one's own community and daily life — are identified and solved. Design thinking is not an academic subject. It is the **spirit of engineering itself** — understanding the human dimension behind every problem.

Whatever the pace of technology or competition, an engineer with genuine **problem-solving ability** remains



Thank
You
♥